

Equilibrium Known-Answer Audit of Cartesian Bath Heat and Medium Entropy

Numerical validation report

1 September 2026

1 Question and frozen protocol

This audit tests whether the bath-heat and medium-entropy accumulators in `NLS_entropy_ft.cpp` have the correct sign and temperature normalization. Internal first-law consistency is not used as the ground truth. Instead, the same Cartesian sampler is run at equilibrium, where $T_L = T_R = T$ and the exact stationary answers are zero mean entropy production, zero mean bath and energy currents, and a distribution of finite- time medium entropy symmetric about zero.

The frozen parameters are $n = 3$, $\gamma = 0.1$, $\Delta t = 5 \times 10^{-4}$, burn-in 500, and block duration $t = 20$. For each of $T = 6$ and $T = 10$, 128 independent streams contribute 7,813 blocks each, for exactly 1,000,064 blocks. Whole-stream bootstrap confidence intervals use 5,000 replicates. The production seeds are 2026090106 and 2026090110, respectively. The full protocol, including the fixed binning and pass/fail rules, was committed before production as `c7e2a92`.

The energy current reported below is derived from the two separately saved bath heats,

$$J_E = \frac{Q_L - Q_R}{2t},$$

and is distinct from the action current.

2 Source-level temperature-factor check

The audited source has SHA-256

9ae5835ed708c8794c8b00ba799b23761482953aaf0ed47cd0b4ba3966d4eaf2

Its boundary update is, verbatim,

```
const double noise_scale = std::sqrt(2.0 * GAMMA * temperature);
state.x[site] += -GAMMA * state.force_real[site] * dt
               + noise_scale * sqrt_dt * normal_x;
state.y[site] += -GAMMA * state.force_imag[site] * dt
               + noise_scale * sqrt_dt * normal_y;
```

where the force is ∇E . Therefore each thermostatted Cartesian component obeys the target continuous SDE

$$dX = -\gamma \partial_X E dt + \sqrt{2\gamma T} dW.$$

The forward bath operator is

$$L_{\text{bath}}^* p = \gamma \nabla \cdot (p \nabla E + T \nabla p),$$

which annihilates $p_{\text{eq}} \propto e^{-E/T}$. Equivalently, $T_{\text{eff}} = (\sqrt{2\gamma T})^2 / (2\gamma) = T$, giving exactly 6 and 10 for the two input temperatures. Thus there is no hidden factor two in the target SDE. The production runs below independently test the finite-step implementation.

3 Zero-mean known answers

Table 1 gives the raw estimates, stream-level standard errors, sigma-from-zero values, and whole-stream bootstrap 95% confidence intervals. No nonzero value is rounded to zero.

Table 1: Equilibrium means. Every confidence interval contains zero.

T	observable	estimate	SE	estimate/SE	CI low	CI high
6	Q_L/t	7.6784422×10^{-4}	1.5975595×10^{-3}	0.4806	$-2.3464676 \times 10^{-3}$	3.8863464×10^{-3}
6	Q_R/t	$-7.5671549 \times 10^{-4}$	1.5967535×10^{-3}	-0.4739	$-3.8776284 \times 10^{-3}$	2.3799250×10^{-3}
6	J_E	7.6227986×10^{-4}	1.5971536×10^{-3}	0.4773	$-2.3507642 \times 10^{-3}$	3.8428207×10^{-3}
6	Σ^m/t	$-1.8547889 \times 10^{-6}$	1.0302400×10^{-6}	-1.8003	$-3.9019577 \times 10^{-6}$	1.3515396×10^{-7}
10	Q_L/t	$-2.4810271 \times 10^{-3}$	3.1458983×10^{-3}	-0.7887	$-8.6265471 \times 10^{-3}$	3.6077970×10^{-3}
10	Q_R/t	2.4822350×10^{-3}	3.1455504×10^{-3}	0.7891	$-3.4715862 \times 10^{-3}$	8.8070629×10^{-3}
10	J_E	$-2.4816311 \times 10^{-3}$	3.1457211×10^{-3}	-0.7889	$-8.6447152 \times 10^{-3}$	3.5888729×10^{-3}
10	Σ^m/t	$-1.2079930 \times 10^{-7}$	8.9844292×10^{-7}	-0.1345	$-1.9280629 \times 10^{-6}$	1.6848915×10^{-6}

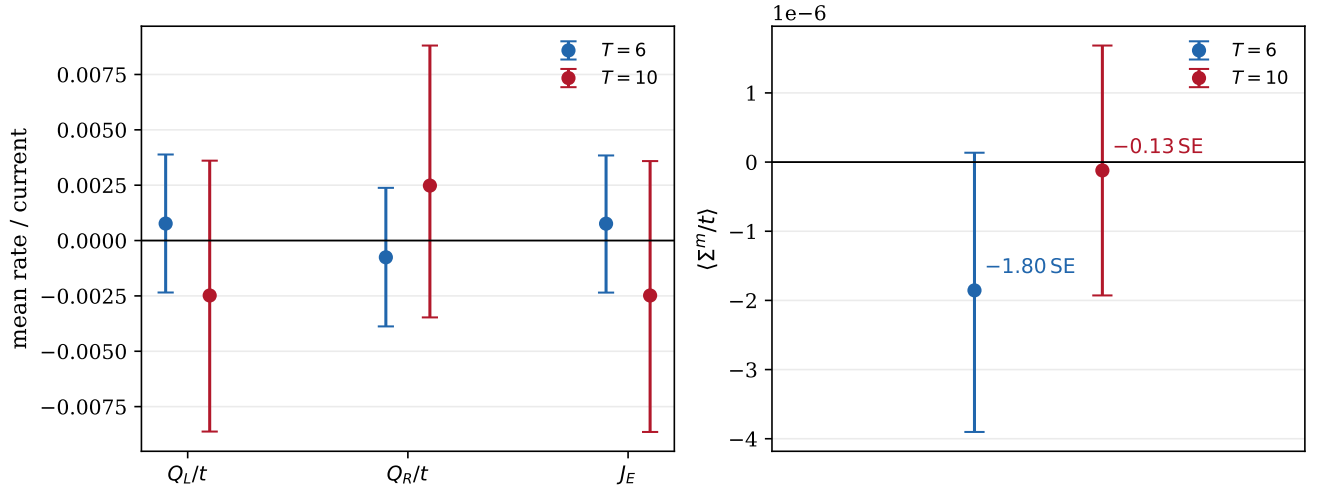


Figure 1: Raw equilibrium means with whole-stream bootstrap 95% confidence intervals. The right panel annotates the entropy estimate in units of its stream-level standard error.

The largest entropy displacement is at $T = 6$, where the estimate is -1.8003 standard errors from zero and the upper CI endpoint is 1.35154×10^{-7} . It therefore passes the predeclared 95%-CI and three-standard-error gates, without being described as numerically equal to zero.

4 Equilibrium symmetry of medium entropy

The raw sign counts are

T	$\#(\Sigma^m > 0)$	$\#(\Sigma^m < 0)$	zeros	counting z	stream z
6	500097	499967	0	0.129996	0.232063
10	499962	500102	0	-0.139996	-0.224329

Both differences are far below one counting standard deviation.

For $a = \Sigma^m/t$, the frozen diagnostic fits

$$\frac{1}{t} \log \frac{p(a)}{p(-a)} = b + s a.$$

At equilibrium the exact slope is $s = 0$, not 1. All 80 symmetric bin pairs pass the predeclared minimum of 200 samples on each side; no fit window is selected after viewing the result.

Table 2: Frozen WLS symmetry fits; confidence intervals are whole-stream bootstrap intervals.

T	slope	bootstrap SE	slope/SE	95% CI	R^2	pairs
6	-1.18034×10^{-4}	1.33429×10^{-3}	-0.0885	$[-2.73956, 2.49981] \times 10^{-3}$	5.31×10^{-5}	80
10	-1.25720×10^{-4}	1.46823×10^{-3}	-0.0856	$[-2.98509, 2.75971] \times 10^{-3}$	7.13×10^{-5}	80

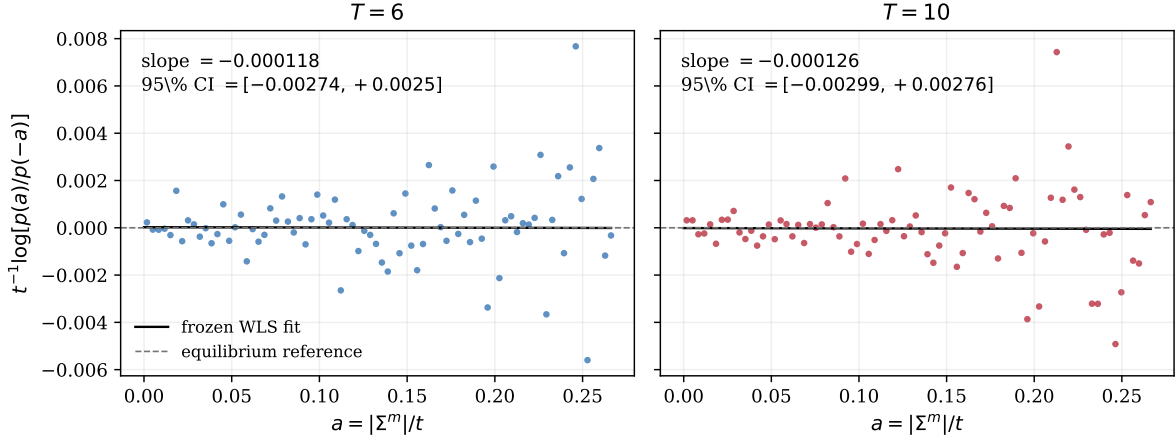


Figure 2: All supported symmetric-bin log-ratios and the predeclared WLS fits. The nearly zero R^2 values are expected under the equilibrium null: there is no systematic linear trend to explain.

5 Integrity, provenance, and verdict

Both compressed block files pass `zstd -t`; each decompresses to 1,000,065 lines (one header plus 1,000,064 data rows). The analysis verifies 128 streams with exactly 7,813 blocks each. Both runs report zero implicit- midpoint failures. The block first-law residual RMS values are 2.36203×10^{-4} at $T = 6$ and 7.43293×10^{-4} at $T = 10$, corresponding to rate RMS values 1.18101×10^{-5} and 3.71647×10^{-5} .

The exact production commands were

```
entropy_ft_n3_eq sample_n3 6 6 3 8 500 20 7813 0.0005
2026090106 8 .../raw/T6 1
entropy_ft_n3_eq sample_n3 10 10 3 8 500 20 7813 0.0005
2026090110 8 .../raw/T10 1
```

The binary SHA-256 is

93c4aa6d046f3c35cd0ea1136091fc44ef1b64baf6237e4663ca9e86b995a156

The compressed raw-data SHA-256 values are 807a4971...54ee22 for $T = 6$ and b5b21983...0e81a8 for $T = 10$; full hashes are in the provenance manifest.

Verdict. The Cartesian heat/entropy definition **passes** the frozen equilibrium known-answer test at both temperatures. These controls rule out the proposed overall sign or factor-two error at the tested parameters and validate the normalization of the driven bath-heat and medium-entropy observables. They do not prove a driven fluctuation theorem, provide a missing system-entropy term, or resolve rare-event sampling limitations.